P.A.S.S. - Password Automation for Surveillance Systems

I. Project Description

The system is a Python-based tool that is designed to automate the management of Hikvision CCTV passwords handled by the ISS department. The system is inspired by the SADP software which is a tool that shows all devices within the same network allowing the administrator to configure selected devices by batch. The proposed system will also allow the administrator to automatically detect Hikvision CCTVs within the same network and perform batch password updates through GUI. The system will not store passwords and will avoid logging or displaying sensitive information.

II. Objectives

- To develop an automatic batch password management tool for Hikvision CCTV devices within the ISS department network.
- To eliminate repetitive manual login processes.
- To reduce the time required to update multiple CCTV passwords.

III. Significance of the Project

The project aims to address the inefficiency of manually changing the password of 300+ CCTVs within the MEC network which the department is required to do biannually. The process of updating passwords of this amount can be tedious and time consuming.

IV. Scope and Limitations

Scope

- Applicable only to Hikvision devices.
- Desktop-based GUI built by Python Tkinter.
- Operates within the same network/VLAN as the CCTVs.

Limitations

- Not compatible with non-Hikvision brands.
- This iteration does not allow managing of passwords of CCTVs in the NVR.

V. Methodology

Phase 1: Core Function Development

In this phase, the functionality of the system will be developed. The project will start with creating the logic for changing passwords through a Python script by sending HTTP requests to the CCTV's URL. A batch processing logic will then be developed to update CCTV passwords sequentially by iterating through the list of detected and selected

CCTV IP addresses. Lastly, an error handling and retry mechanism will be implemented so that failed updates can be identified and handled without interrupting the entire batch process.

Phase 2: GUI Development

During this phase, a GUI will be designed using the Python Tkinter library. The interface will allow administrators to view CCTVs detected within the same network or VLAN and select multiple devices through checkboxes. Input fields will be provided for the username, current password, and new password, along with a start button to initiate the update process. The GUI will also include a results log panel that displays the success or failure status for each camera and a progress counter indicating how many devices have been processed. The interface will be designed to resemble familiar device management tools such as the SADP to make it intuitive for the administrators.

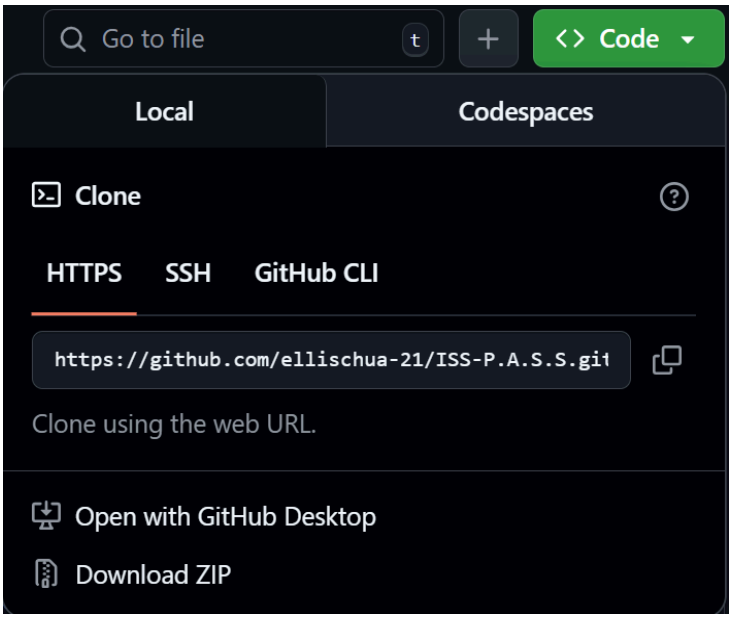
Phase 3: Testing and Validation

In this phase the system will undergo testing to ensure compatibility with different Hikvision camera models and firmware versions. This phase will verify that password updates are correctly applied to selected devices and that error conditions such as incorrect credentials and password policy violations are handled properly.

VI. Walkthrough and Features

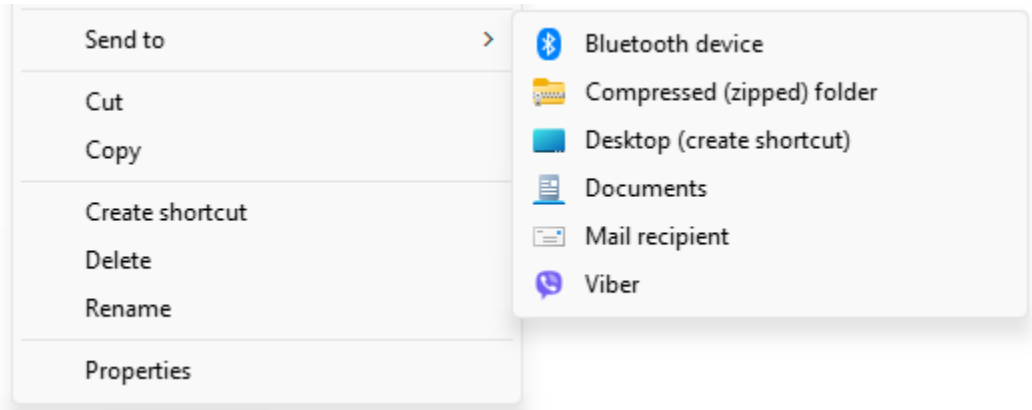
1. Downloading the Program

Access the github link: P.A.S.S. and download the .zip file by clicking on **'Code'** then **'Download ZIP'**. Extract the folder to your chosen directory when download has finished.



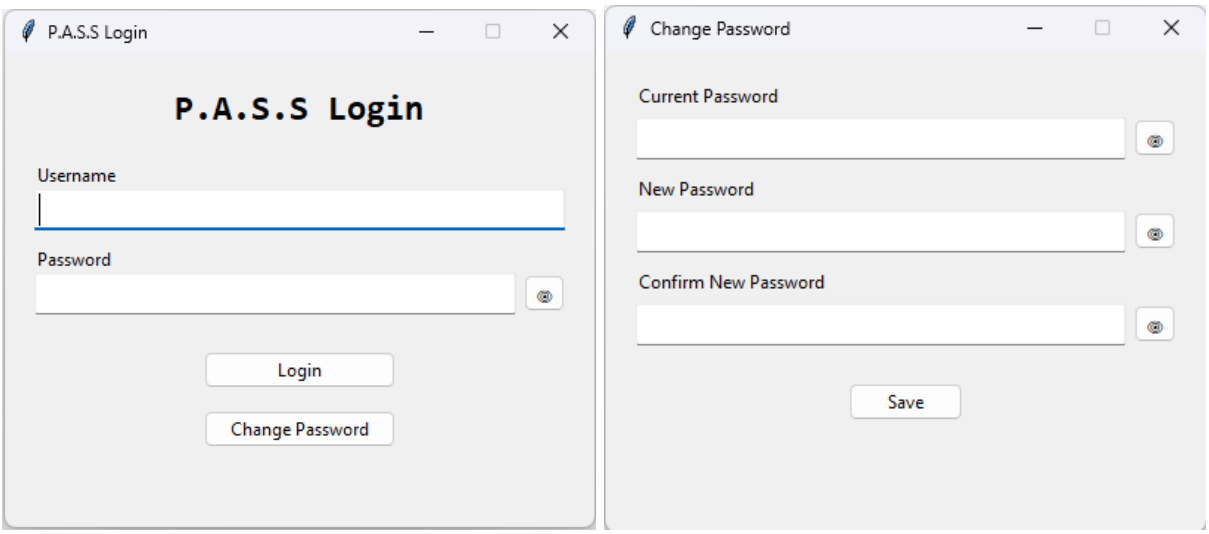
2. Accessing the Program

Open the **'ISS-P.A.S.S.'** folder then go to **'dist'** and double click on the **'P.A.S.S.'** icon to start the program. To make a desktop shortcut, left click on the **'P.A.S.S.'** icon once and then right click, select **'Show more options'**, click **'Send to'** then **'Desktop (create shortcut)'**. <u>It is recommended that the user secures the ISS-P.A.S.S. folder.</u>



3. Logging In and Changing User Password

<u>Before logging in, MAKE SURE to change the user password first.</u> Click on the **'Change Password'** button which will redirect you to a new window. Input the default password *'password123'* and the new password. Click **'Save'**. To log in, simply input the username and the new password then press **Enter** on your keyboard or click the **'Login'** button.



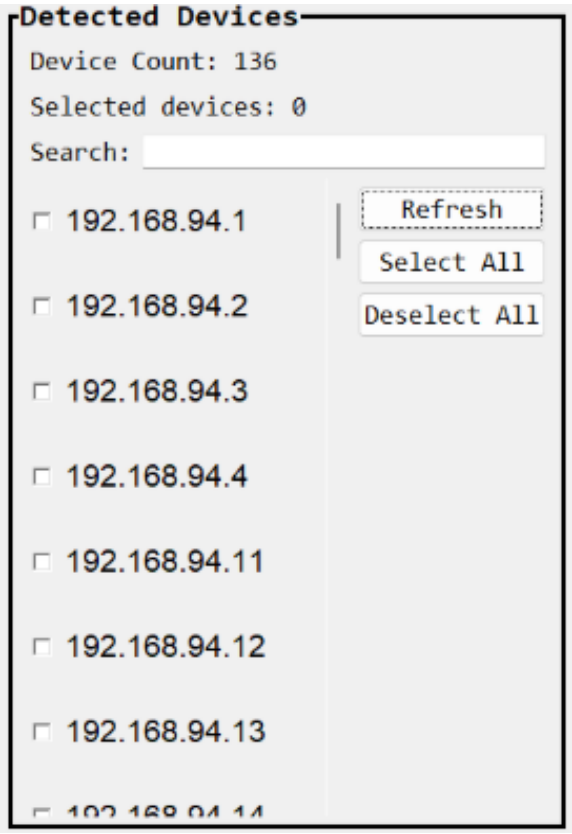
Forgot Password?

In case of forgetting the password, the user can open the **'ISS-P.A.S.S.'** folder, go to **'dist'**, then delete the **'login_config.json'** file. The password will revert back to the default password and the user can apply a new password.

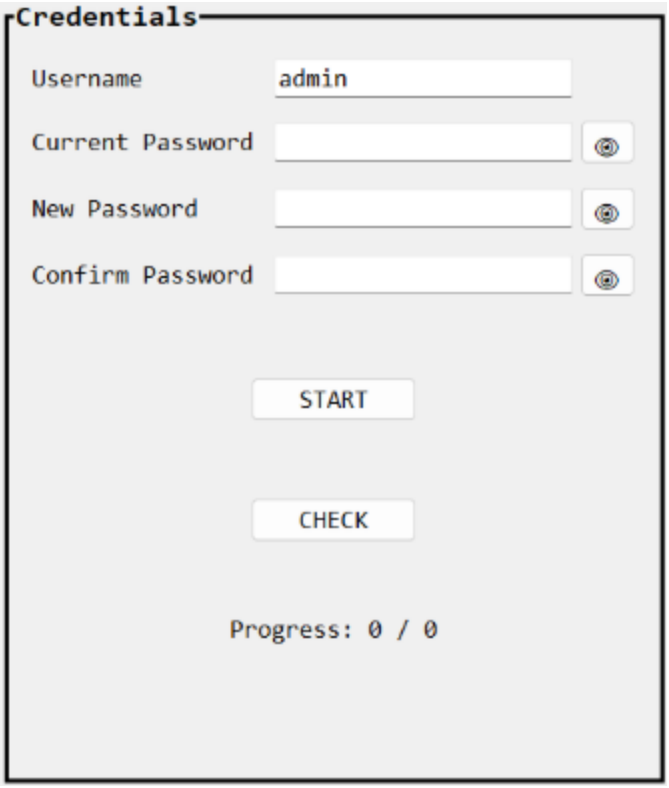
4. Main GUI

Upon login, the user will be greeted with the Main GUI which is divided into three sections:

Detected Devices Frame - Responsible for the displaying, navigation, and selection of Hikvision devices detected in a specific VLAN. It contains a scrollable checkbox list of the detected device IPs and features such as labels showing the total number of detected devices and the number of selected devices as well as a search bar for quick finding of specific device IPs. It also contains buttons for batch selection/deselection and refreshing that repeats the scanning of the VLAN which is useful when changing VLANs or confirming that all devices are already scanned.



Credentials Frame - Responsible for managing the password changing process. It contains basic features such as the Username and Current, New, and Confirm Password fields. The **'START'** button initiates the process of changing passwords and the **'CHECK'** button lists all selected IPs in the Results Log Frame for double checking.



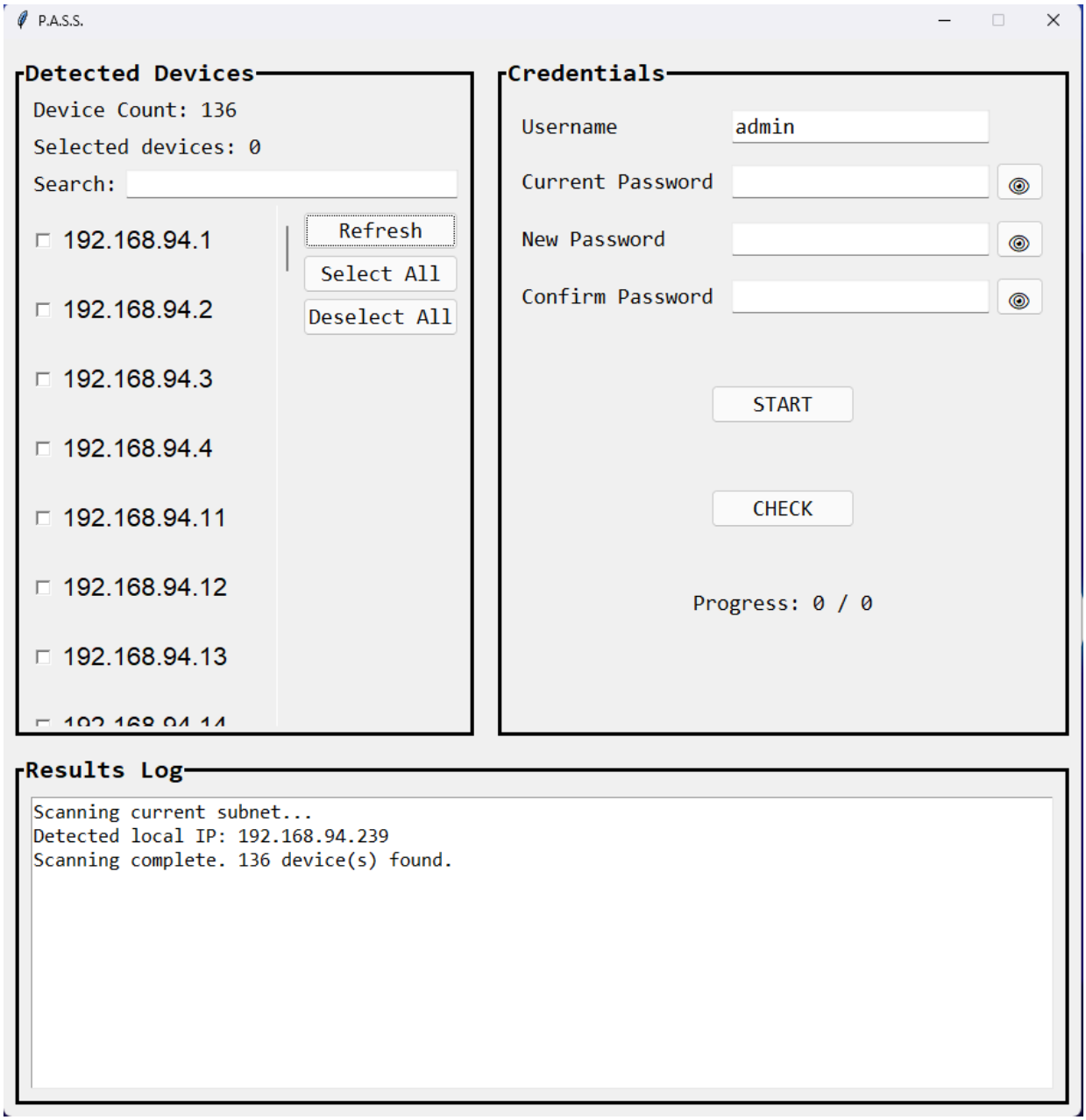
Results Log Frame - Responsible for displaying updates for events, actions, or errors that provides users visibility with their interactions.

VII. How to Change CCTV Password

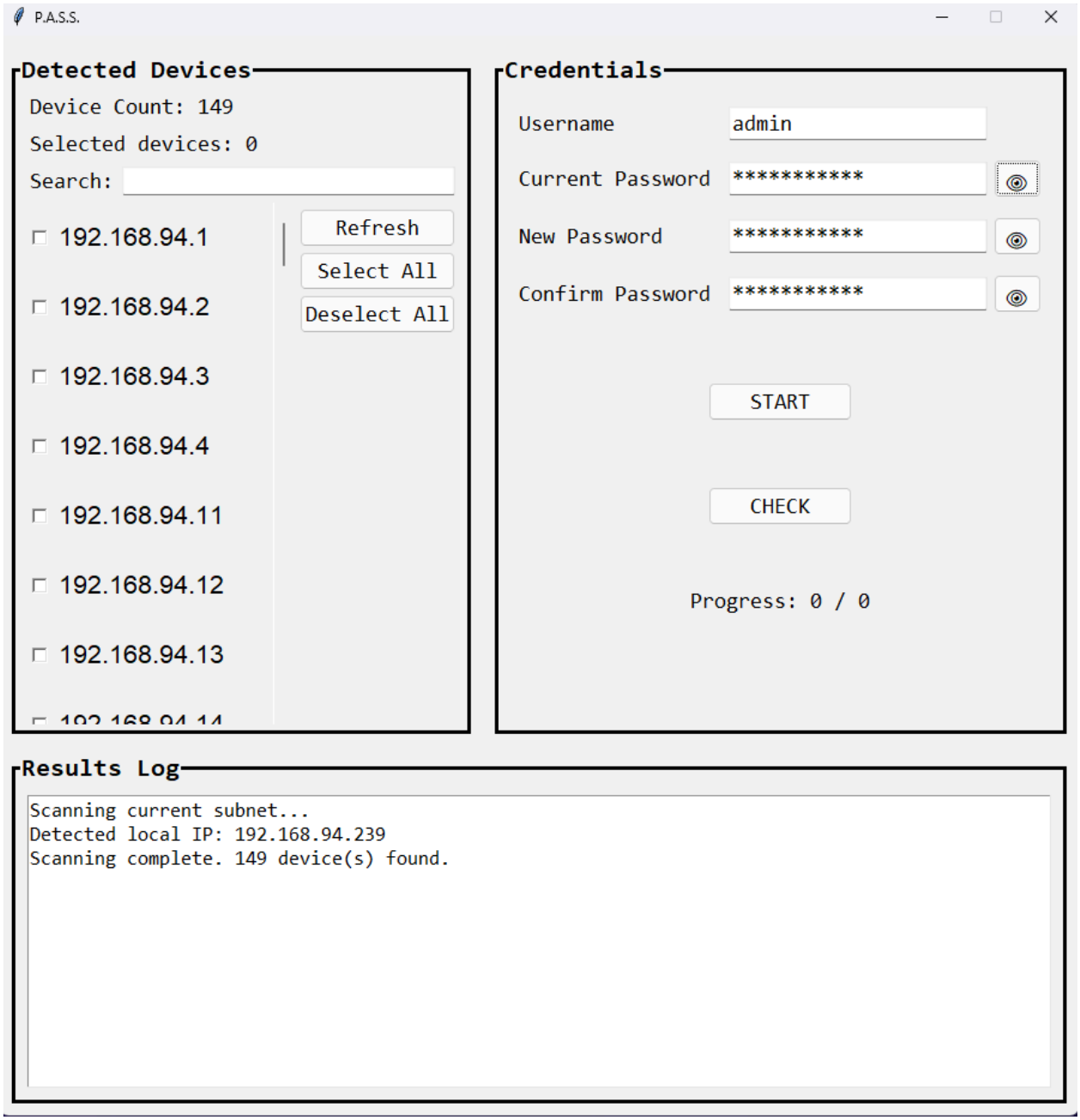
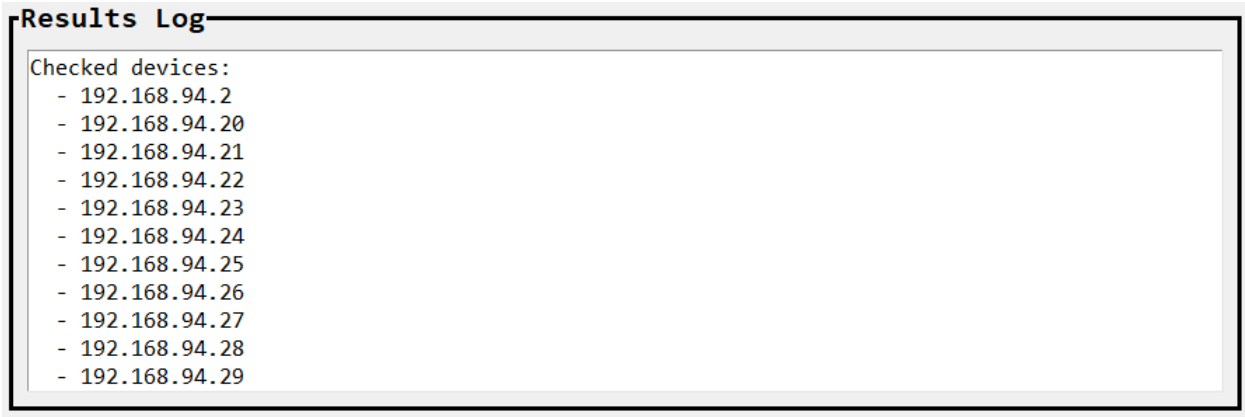
First, set your device IP address to have the same network ID with the VLAN of the devices you wish to change passwords. Make sure to avoid setting the user's IP address to an existing one. Connect to the network via Ethernet and start the program.

Upon starting, the program will start with scanning the VLAN for Hikvision devices based on the IP address set on the user's device. This will be reflected in the results log where it displays that it has successfully defected the user device IP address. In the example screenshot below, the user device IP address is set to 192.168.94.239, therefore, the program will scan for devices in VLAN 94.

After successfully scanning, a checkbox list of all detected devices will appear on the **'Detected Devices Frame'**. To make sure that all devices are detected, the user can **'Refresh'** to double check.



Select all the IP addresses of the devices whose passwords the user wants to change by simply checking them. Enter the necessary credentials in the **'Credentials Frame'**. Follow Hikvision's password protocol of 8 character minimum length and must contain at least 1 number. By default, Hikvision devices are set to **'admin'** so the program autofills it already. In cases that the device username is not set to the default, simply input the new username.



Before pressing the **'START'** button, it is recommended to first press the **'CHECK'** button. This feature will log all checked IP addresses in the 'Results Log Frame' and will allow the user to double check. Proceed with the password changing process by pressing the **'START'** button and clicking yes on the confirmation prompt.